Visual Odometry

2026-03-27

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Visual odometry

Visual SLAM

- Simultaneous
- Localization
- And
- Mapping

Image

Features

- SIFT
- ORB

Easily identifiable points

- finger print descriptor

Pinhole camera model

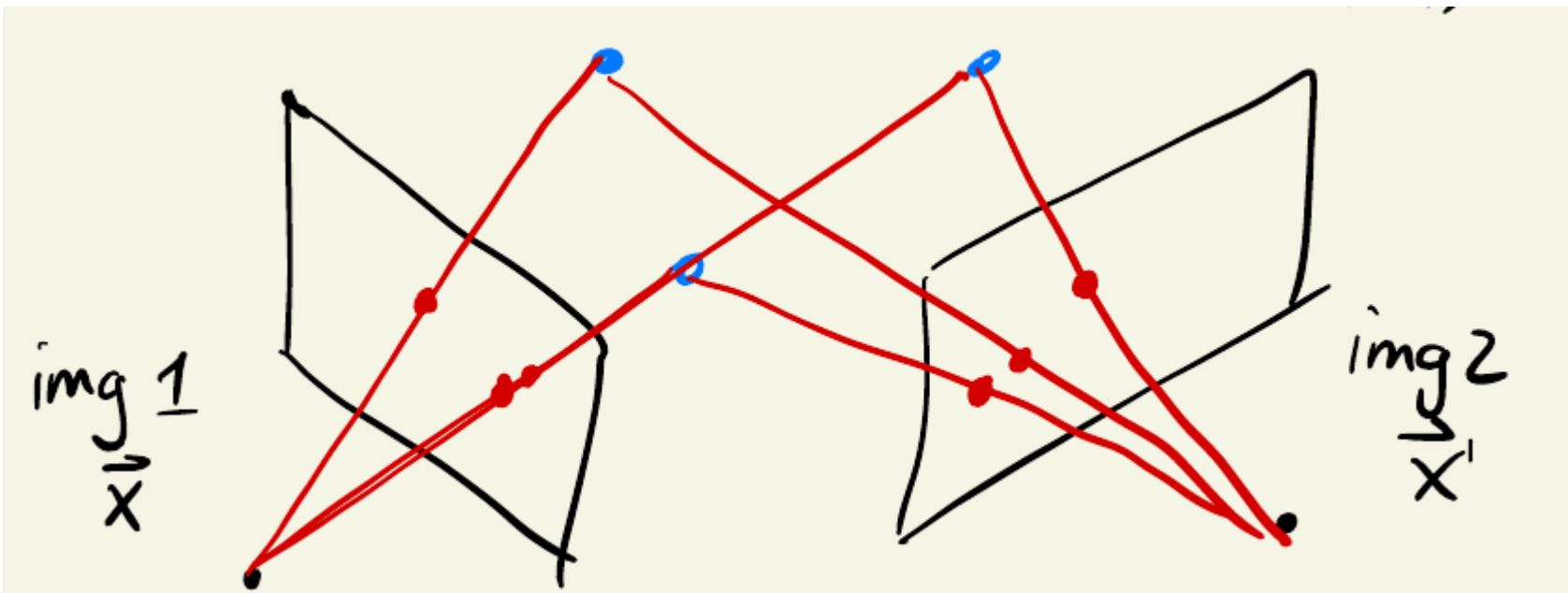
The pinhole camera model is given by

$$\vec{x} = K \cdot [Rt] \cdot \vec{X}$$

And here with all the elements expanded

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & \alpha & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Pair of images - Fundamental matrix



Fundamental matrix

$$\vec{x}_1^T \cdot F \cdot \vec{x}_2 = 0$$

Where $\vec{x}_1$ is a point in image one and $\vec{x}_2$ is a point in image two. $\vec{x}_1$ and $\vec{x}_2$ are given in homogeneous coordinates.

Fundamental matrix - size of elements

The size of the elements in the equation is as follows

$$< 3 \times 1 >^T \cdot < 3 \times 3 > \cdot < 3 \times 1 >$$

Normalized image coordinates

To simplify the following calculations, the <u>normalized image coordinates</u> ($\vec{y}$) are introduced.

The normalized image coordinates are related to the actual image coordinates $\vec{x}$ through the relation.

$$\vec{x} = K \cdot \vec{y}$$

Normalized image coordinates (2)

Where K is the camera matrix

$$K = \begin{pmatrix} f_x & \alpha & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

The normalized image coordinates can therefore be determined as this

$$\vec{y} = K^{-1} \cdot \vec{x}$$

Essential matrix

Using normalized coordinates the essential matrix fulfills this equation for all corresponding points between two images.

$$\vec{y}_1^T \cdot E \cdot \vec{y}_2 = 0$$

Property of the essential matrix

Consider how the point $\vec{Q}$ is seem by two different cameras.

The first camera is placed at the origin of the coordinate system, whereas the second camera is placed at $\vec{t}$ with an orientation described by the rotation matrix R.

The normalized image coordinates of the point $\vec{Q}$ in the two cameras are then:

$$\begin{aligned}\vec{y}_1 &= I \cdot (\vec{Q} - \vec{0}) \\ &= \vec{Q} \\ \vec{y}_2 &= R \cdot (\vec{Q} - \vec{t})\end{aligned}$$

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Math properties

A few properties to use in the following calculations

$$R^T \cdot R = R \cdot R^T = I$$

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0$$

Cross product in matrix form

The value $[t]_x$ has the property

$$[t]_x \cdot \vec{a} = \vec{t} \times \vec{a} = \begin{pmatrix} t_2 a_3 - t_3 a_2 \\ t_3 a_1 - t_1 a_3 \\ t_1 a_2 - t_2 a_1 \end{pmatrix}$$

From this the content of $[t]_x$ can be found to be

$$[t]_x = \begin{pmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{pmatrix}$$

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Hidden information

$$E = R \cdot [t]_x$$

Property of the essential matrix

$$
\begin{aligned}
\vec{y}_2^\top \cdot E \cdot \vec{y}_1 &= \vec{y}_2^\top \cdot R \cdot [t]_x \cdot \vec{y}_1 \\
&= (R(\vec{Q} - \vec{t}))^\top \cdot R \cdot [t]_x \cdot \vec{Q} \\
&= (\vec{Q} - \vec{t})^\top \cdot R^\top \cdot R \cdot [t]_x \cdot \vec{Q} \\
&= (\vec{Q} - t)^\top \cdot [t]_x \cdot \vec{Q} \\
&= \vec{Q}^\top \cdot [t]_x \cdot \vec{Q} - \vec{t}^\top \cdot [t]_x \cdot \vec{Q}
\end{aligned}
$$

Property of the essential matrix

$$
\begin{aligned}
\vec{y}_2^\top \cdot E \cdot \vec{y}_1 &= \vec{Q}^\top \cdot [t]_x \cdot \vec{Q} - \vec{t}^\top \cdot [t]_x \cdot \vec{Q} \\
&= \left([t]_x^\top \cdot \vec{Q}\right)^\top \cdot \vec{Q} - \left([t]_x^\top \cdot \vec{t}\right)^\top \cdot \vec{Q} \\
&= \left(-[t]_x \cdot \vec{Q}\right)^\top \cdot \vec{Q} - \left(-[t]_x \cdot \vec{t}\right)^\top \cdot \vec{Q} \\
&= \left(-\vec{t} \times \vec{Q}\right)^\top \cdot \vec{Q} + \left(\vec{t} \times \vec{t}\right)^\top \cdot \vec{Q} \\
&= \left(-\vec{t} \times \vec{Q}\right)^\top \cdot \vec{Q} + \left(\vec{0}\right)^\top \cdot \vec{Q} \\
&= 0
\end{aligned}
$$

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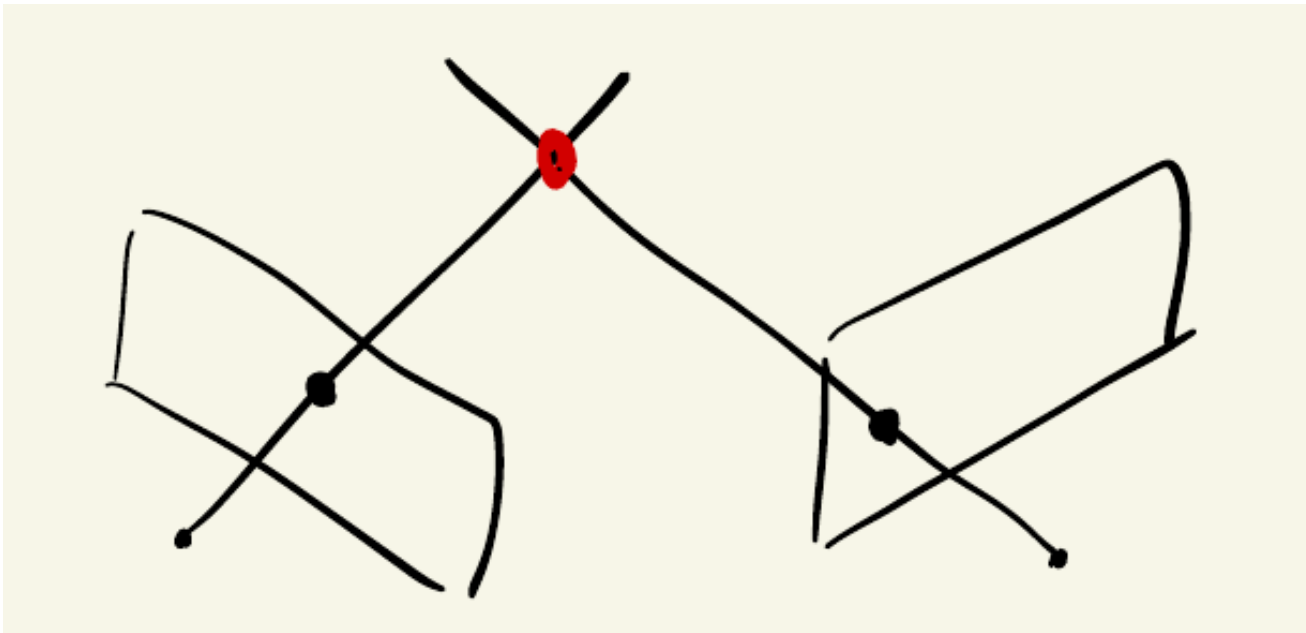
Triangulate 3D points



Given

- 2 camera poses
- 2 pairs of image coordinates

Triangulate 3D points - mathematics

The point $\vec{Q}$ has been observed by two cameras. The first camera placed at $[R\ t]_1$ has seen $\vec{Q}$ at the image coordinate $\vec{x}_1$. The second camera placed at $[R\ t]_2$ has seen $\vec{Q}$ at the image coordinate $\vec{x}_2$.

$$\vec{x}_1 = K \cdot [R\ t]_1 \cdot \vec{Q}$$
$$\vec{x}_2 = K \cdot [R\ t]_2 \cdot \vec{Q}$$

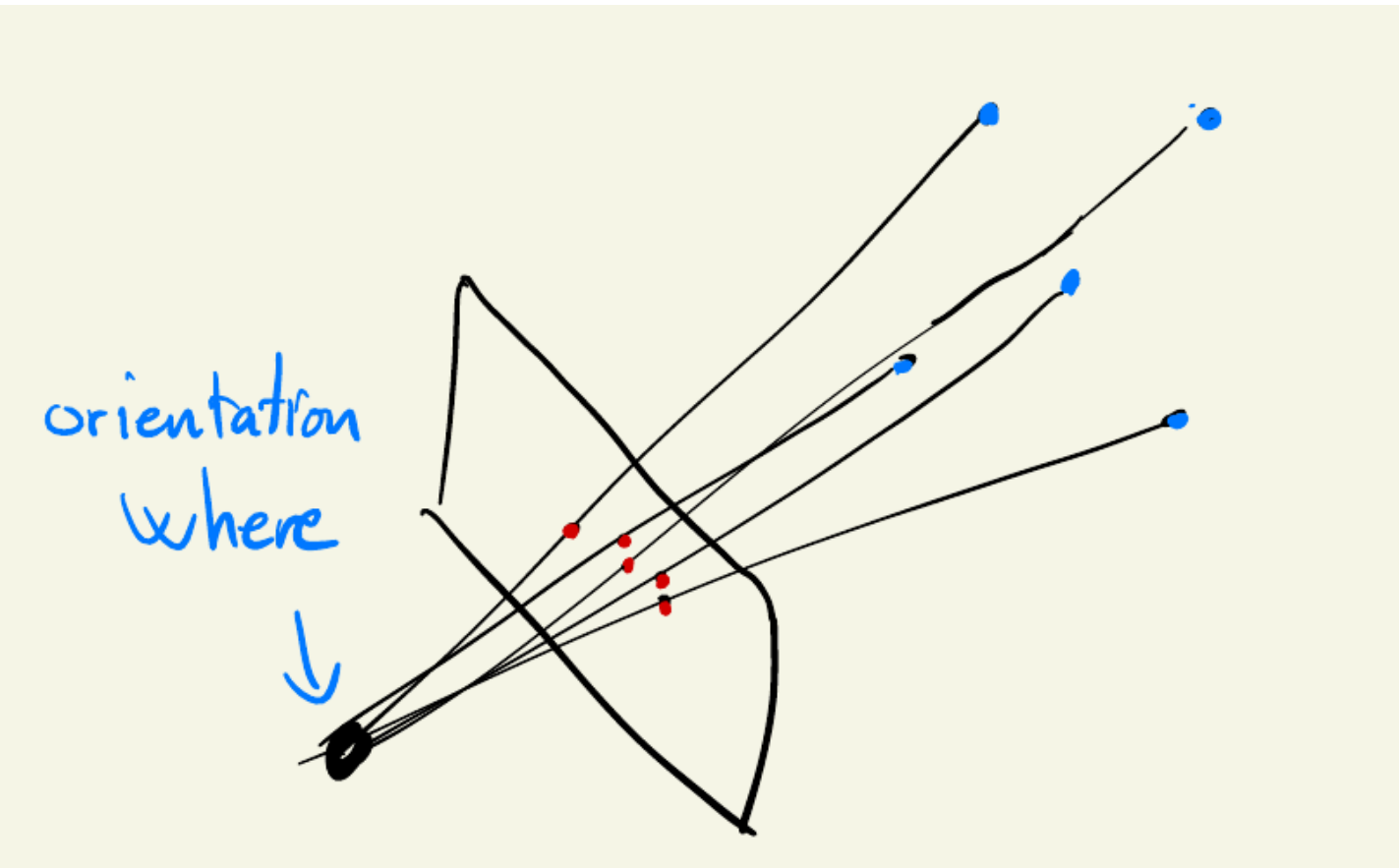
Notice

Cameras measure angles

We can only determine direction of motion, not distance.

Camera positioning

Given a set of 3D points and their location in an image the camera location can be determined.



Visual SLAM

Algorithm initialization

Map initialization

- Given two images
- match features
- estimate camera motion
- 3D point reconstruction (with associated feature descriptors)
  - Following bundle adjustment refinement

Algorithm main loop

For new features

- match features with prev. frame
- match features with map
- estimate camera location*
- add 3D points to map from triangulation with previous frame*

* bundle adjustment

Minimization of reprojection error

Error reduction

- bundle adjustment
  - g20 library
- minimize reprojection error

Ransac

Ransac general algorithm

Ransac: Random Sample Consensus

```
Repeat N times
- Sample k points
- Fit model to sampled points
- Count number of inliers
```

Ransac line detector

Ransac: Random Sample Consensus

```
Repeat N times
- Sample 2 points
- Fit line to sampled points
- Count number of inliers
```

Ransac video

Dealing with Outliers: RANSAC | Image Stitching

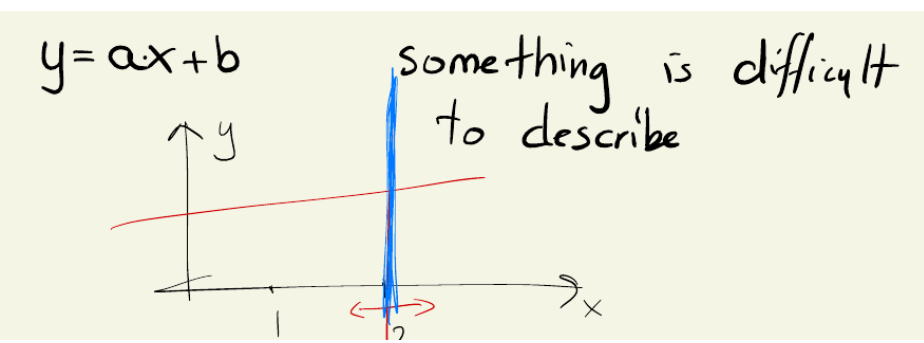
First Principles of Computer Vision

Watch on

Operations on points and lines

Points and lines in homogeneous coordinates

Using the equation $y = a \cdot x + b$ some lines can be described easily, whereas others are more difficult to represent.



This representation is more robust:

$$a \cdot x + b \cdot y + c \cdot 1 = 0$$

Comparing points and lines

Point: $(x_0, y_0, 1) = \vec{x}$

Line: $(a, b, c) = \vec{l}$

If the point $\vec{x}$ lines on the line $\vec{l}$ this holds

$$\vec{l}^T \cdot \vec{x} = 0$$

The signed distance between the point and the line can be calculated as follows

$$d = \frac{\vec{l}^T \cdot \vec{x}}{\sqrt{a^2 + b^2}}$$

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Intersection of two lines

Given the two lines $\vec{l}_1$ and $\vec{l}_2$, their intersection can be found by calculating their cross product.

$$\vec{x} = \vec{l}_1 \times \vec{l}_2$$

We can calculate the distance between the calculated intersection and the two lines.

$$\vec{l}_1^T \cdot \vec{x} = \vec{l}_1^T \cdot \vec{l}_1 \times \vec{l}_2 = 0$$
$$\vec{l}_2^T \cdot \vec{x} = \vec{l}_2^T \cdot \vec{l}_1 \times \vec{l}_2 = 0$$

Line between two points

Similarly if we are given two points $\vec{x}_1$ and $\vec{x}_2$, the line that connects them can be found as follows:

$$\vec{l} = \vec{x}_1 \times \vec{x}_2$$

We can calculate the distance between the calculated line and the two points.

$$\vec{x}_1^T \cdot \vec{l} = \vec{x}_1^T \cdot \vec{x}_1 \times \vec{x}_2 = 0$$
$$\vec{x}_2^T \cdot \vec{l} = \vec{x}_2^T \cdot \vec{x}_1 \times \vec{x}_2 = 0$$

Outlier detection

Assume that we have found two matching image features (at least according to their feature descriptors). How can we verify that it is a real match and not a coincidence?

Look at the condition from the fundamental matrix.

$$\vec{x}_1^T \cdot F \cdot \vec{x}_2^T = 0$$

$$\vec{x}_1^T \cdot F = \vec{l}$$

$$d = \frac{\vec{l} \cdot \vec{x}_2}{\sqrt{a_1^2 + b_a^2}}$$

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Data structures

Data structures 1

Map

- 3D point and features

Camera poses

- Where have each camera been

Feature locations

- where have each feature been seen

Data structures 2

TrackedPoint

- Spatial coordinate (3D)
- feature descriptor
- point_id

TrackedCamera

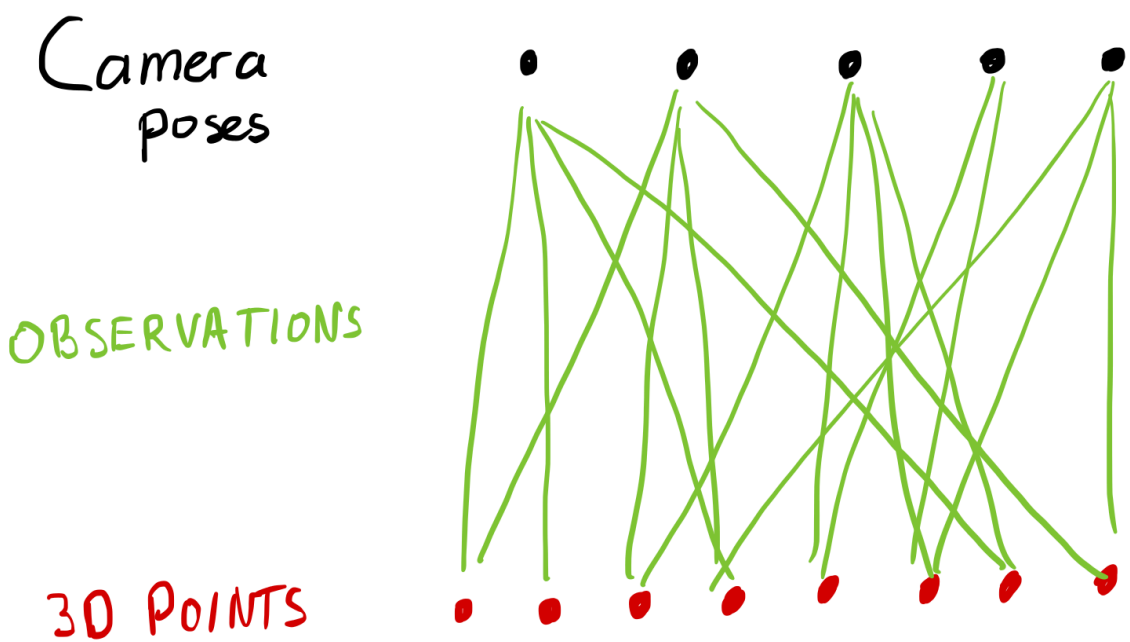
Data structures 3

Observation

- point_id
- camera_id
- image_coordinates

Scaling to many images

Using g2o

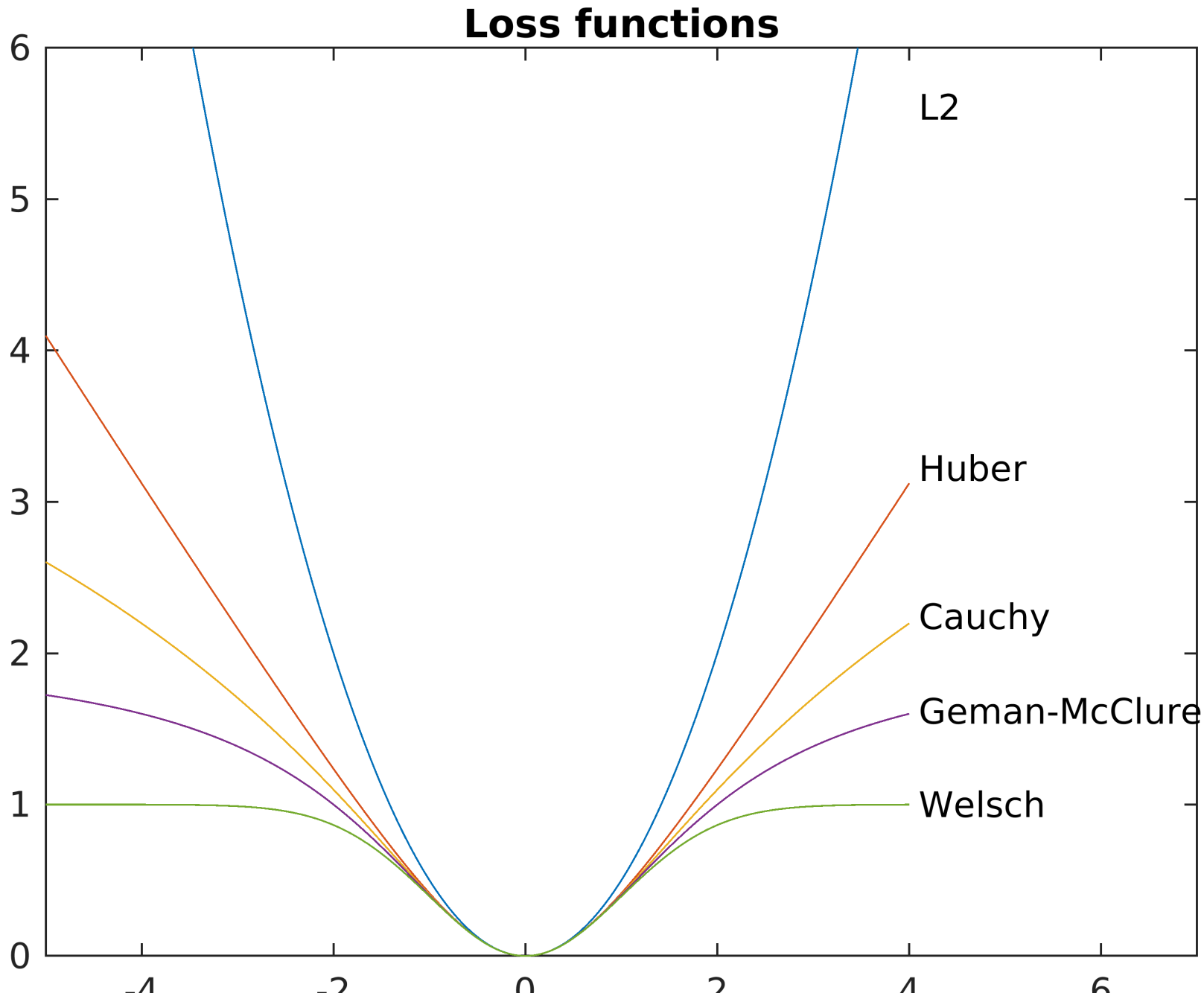


Camera poses (vertices)

3D points (vertices)

Observations (edges)

Robust loss functions


Loss functions
L2
Huber
Cauchy
Geman-McClure
Welsch
6
5
4
3
2
1
0
-4
-2
0
2
4
6

Outro

Course overview

2026-02-01

- Python and 2D image processing
- Camera in space
- Optimization
- Fiducial markers

2026-03-01

- Camera calibration mini project
- Feature detectors and video stabilization
- Motion detection
- Visual odometry

2026-04-01

- Visual odometry mini project
- Orthomosaics
- Pumpkin counting mini project

2026-05-01

- Bag of Visual Words / AI overview

Quote

“Of course, it is very important to be sober when you take an exam. Many worthwhile careers in the street-cleansing, fruit-picking and subway-guitar-playing industries have been founded on a lack of understanding of this simple fact.”

— Terry Pratchett, Moving Pictures